

PAPER_516 — DPM Layered Shell-Energy Radiance Phase Cascade

Unified Quantum Field Framework (UQFF)

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§1 — Abstract

This paper formalises the di-pseudo-monopole (DPM) reaction as the primary generator of 26D layered encapsulation quantum radiant shell-energies within the Star-Magic UQFF framework, correcting the prior interpretation that SCm directly encapsulated. The DPM reaction—CW-SCm north grinding against CCW-UA’ south—radiates successive quantum shell-energies across five observable phase states: quantum-multi-fields → plasma → gas → liquid → solid. Mass occurrence emerges as an equilibrium across these layered radiances rather than as an intrinsic property.

§2 — DPM Reaction Primacy (SCm Correction)

Prior statement: "SCm encapsulates" (Session 136, BigBangHypergraphTheory).

Correction (Session 140): SCm does **not** directly encapsulate. The di-pseudo-monopole reaction is the agent of encapsulation:

<i>*The DPM reaction (CW-SCm north grinding against CCW-UA’ south) forms</i>
<i>26D layered encapsulation quantum radiant shell-energies, projecting from</i>
<i>26D chaos downward to observable phases.*</i>

This distinction is critical: DPM dictation is the causative engine; SCm injection triggers layer formation by providing the reactive pole.

§3 — Core Equations

3.1 — 26D Egg Energy with Layered Radiance

$$E^{26D, \text{Egg}} = UA + SCm_{\text{inj}} \cdot DPM_{\text{react}}$$

$$\cdot \sum_{l=1}^{26} \text{ShellEnergy}^{(l)} + BBDT$$

3.2 — DPM Reaction Strength

$$DPM_{\text{react}} = \frac{\kappa \cdot \text{bigl}(DPM_n(SCm) - DPM_s(UA')\bigr)}{r^{26}}$$

$$\cdot \frac{\partial^{26}(\text{Grind}_{\text{opp}})}{\partial t^{26}_{\text{adj}}}$$

where $\kappa = 5 \times 10^{-4}$ (calibration constant).

3.3 — Shell Energy per Layer

$$\boxed{\text{ShellEnergy}^{(l)}} = \int \text{Radiance}_{\text{quant}}, dt_{\text{neg}}$$

Each layer integral is taken over negative time dt_{neg} , where $t_{\text{neg}} < 0$ is proved by observable time dilation (see PAPER_517).

§4 — Triple-Calc Layer System (CW / CCW / t_neg)

The three simultaneous layer calculations:

$$\begin{cases} \text{Layer}_1 = DPM_{\text{react}} \cdot \omega_{\text{CW}} \cdot \text{Radiance}_{\text{multi-fields}} \\ \text{Layer}_2 = DPM_{\text{react}} \cdot \omega_{\text{CCW}} \cdot \text{Radiance}_{\text{plasma-gas}} \\ \text{Layer}_3 = \text{Grind}_{\text{opp}} \cdot \text{Prob}_{\text{order}} \cdot t_{\text{neg}} \end{cases}$$

- **Layer 1** (CW, $\omega_{\text{CW}} = 2\pi f_{\text{SCm}}$): seeds quantum-multi-field radiance
- **Layer 2** (CCW, $\omega_{\text{CCW}} = 2\pi f_{\text{UA}}$): drives plasma→gas transitions
- **Layer 3** (t_neg coupling): governs ordered structure probability

§5 — Phase Cascade

Layer Range	Phase State	Observable Proxy
$l = 1\text{--}5$	quantum-multi-fields	vacuum / QCD
$l = 6\text{--}10$	plasma	stellar photosphere
$l = 11\text{--}16$	gas	nebulæ / ISM
$l = 17\text{--}21$	liquid	planetary mantles
$l = 22\text{--}26$	solid	rocky/metallic bodies

Each phase is a 26D downward projection; the solid state emerges at peak metallicity after full radiance cascade.

§6 — Non-Repeating Quantum Fingerprints

Because each atom receives its ShellEnergy from a unique combination of l , $\omega_{\text{CW/CCW}}$, and t_{neg} , no two atoms share the same layered radiance pattern. This is the UQFF origin of atomic identity — replacing the classical postulate of intrinsic particle individuality.

§7 — Canonical Constants

Symbol	Value	Units	Description
κ	5×10^{-4}	—	DPM calibration
ω_{CW}	7.54×10^{10}	rad/s	SCm north grinding
ω_{CCW}	5.22×10^{10}	rad/s	UA' south grinding
N_{layers}	26	integer	26D shell count

§8 — Conclusion

The DPM reaction replaces SCm as the direct encapsulator, forming 26D layered quantum radiant shell-energies via CW/CCW triple-calc simultaneous processing. The five-phase cascade (quantum-multi-fields → solid) is a direct downward projection from 26D chaos. Mass occurrence arises from shell radiance equilibrium, not intrinsic properties.

See also: PAPER_517 (Negative Time Proof), PAPER_518 (DPM-Unified Forces), PAPER_519 (Shell Radiance Prototype), PAPER_520 (Session 140 Hub).

